INCH-POUND
MIL-H-21660A(AS)
30 September 1992
SUPERSEDING
MIL-H-21660(Aer)
21 OCTOBER 1958

MILITARY SPECIFICATION

HOOK, THRUST, JATO, 1000 POUNDS NOMINAL THRUST

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the manufacture, inspection, and testing requirements for hook, thrust, Jet Assisted Take-Off (JATO), 1000 pounds nominal thrust.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2b).

SPECIFICATIONS

FEDERAL

A-A-1898	Cushioning Material, Cellulosic, Packaging
L-P-378	Plastic, Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department (Code SR3) Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-J-8667	Jet-Assisted Take-Off Systems Design and Installation of; in Naval Aircraft
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAF 400095	Hook, Thrust - JATO; 1000 Pounds Nominal Thrust
NAF 49489	Jig

(Unless otherwise indicated, copies of drawings are available from the Naval Air Technical Services Facility (NATSF) (Code 3121), 700 Robbins Avenue, Philadelphia, PA 19111-5087.))

* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2b).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117 Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2c), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials. Material shall conform to the specifications as specified herein and on the drawings. Materials which are not covered by specifications shall be of the best commercial quality, of the lightest practical weight, and entirely acceptable for the purpose intended.

3.2.1 Selection of specifications and standards. Specifications and standards for all necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970 except as provided in the following paragraph.

3.2.1.1 AN or MIL standard parts. AN or MIL standard parts shall be used and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, may be used, provided they possess acceptable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and on the contractor's drawings. If there is no acceptable corresponding AN or MIL standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

* 3.3 Manufacturing. The thrust hooks shall be manufactured in accordance with NAF Drawing 400095.

* 3.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and physically interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-STD-100.

3.5 Performance. Hooks shall function under all tests specified in section 4.

3.6 Workmanship. The workmanship shall be that required by the best commercial practice, covering this type equipment, for the quantity production of analogous parts. All operating parts of the hooks shall be free from imperfections which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

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* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table I.

* 4.3.1 First article samples. Unless otherwise specified, after award of a contract or order, the first three hooks shall be delivered for inspection and testing as directed by the acquiring activity (see 6.2e and 6.3).

4.3.2 Correction of defects. Any defects discovered during the inspection and testing of the first article test samples shall be corrected by the contractor. Such measures to correct these defects shall likewise be incorporated in all subsequent hooks offered for acceptance under contract.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following examinations and test and as specified in table III.

a. Examination of product (see 4.5.1).

b. Magnaflux inspection (see 4.5.2).

c. Sampling inspection (see 4.4.1).

* 4.4.1 Sampling inspection. Samples shall be selected by the Government inspector at random, from each production lot, in accordance with table II. Samples shall be forwarded to a test activity designated by the acquiring activity (see 6.2e).

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TABLE I. First article inspection.

Inspection	Test method	Sample no. 1	Sample no. 2	Sample no. 3
Physical characteristics	4.6.1		All	
Salt spray test	4.6.2			X
Vibration test	4.6.3	X		
Release function test	4.6.4		All	
Life test	4.6.5		X	X
Static and repeat load tests	4.6.6	X		X
Destruction test	4.6.7	X		X

Table II. Quality conformance samples.

Total no. of hooks 1/	No. of samples selected
99 or less	2
100 to 299	3
300 to 500	4

1/ When the lot size exceeds 500 hooks, the lot shall be subdivided into lots of of 500 units and the selection for each subdivision shall be in accordance with the above table.

TABLE III. Quality conformance inspection.

Inspection	Test method	Sample size
Physical characteristics	4.6.1	All
Release function test	4.6.4	All
Life test	4.6.5	All
Static and repeat load tests	4.6.6	All
Destruction test	4.6.7	All

4.5 Inspection methods.

4.5.1 Examination of product. Each hook submitted for acceptance under contract shall be inspected visually to ensure that it has been properly assembled and that the workmanship and the finishes are in accordance with the provisions of this specification.

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4.5.2 Magnaflux inspection. After machining and prior to plating, each hook shall be subjected to a magnaflux inspection. The magnaflux inspection shall be repeated after plating (see 6.3).

4.5.3 Commercial materials. When materials are procured under commercially recognized non-Governmental specifications, such as SAE numbers, a certificate showing compliance with such specifications shall be furnished by the supplier of the material (see 6.3).

4.6 Test methods.

4.6.1 Physical characteristics. The units submitted for this test shall be photographed if necessary to illustrate clearly appearance and general arrangement. The units shall also be weighed and subjected to a dimensional, material, and a process check.

4.6.2 Salt spray test. Salt spray test shall be made in accordance with ASTM B 117.

4.6.3 Vibration test. Test hooks shall be mounted in a slosh vibration machine so that the longitudinal axis of the mockup JATO bottle is at right angle to the axis of rotation of slosh machine. JATO hooks shall be vibrated twenty-five (25) hours as follows:

1. Vibration frequency - 1700 cycles per minute (CPM)
2. Vibration amplitude - 0.032 inch

Any undue wear of hook components or release of bottle during test period due to vibration, mechanical failure, or loosening of parts shall be considered cause for rejection of the test hook.

4.6.4 Release function test. A test apparatus suitable for simulating the jettisoning of a 141 pound mock JATO bottle shall be prepared. The hooks shall be so mounted on the apparatus that the force required to jettison the unit shall produce a torque of 150 inch-pounds minimum and 300 inch-pounds maximum on the jettison lever in accordance with MIL-J-8667 for type I JATO hooks. Ten consecutive operations resulting in complete jettisoning of the bottle shall comprise a successful test. Each unit manufactured shall be so tested.

* 4.6.5 Life test. The life test fixture shall be constructed such that the test hooks are held in a horizontal position. The test hooks shall then be subjected to 500 releases. The test shall be conducted at room temperature and a mock JATO bottle of 141 pounds used throughout the test. Record the efficiency and percentage of total retraction between the 1st and 10th releases, inclusive, and the 491st to 500th, inclusive. If the performance of the test hooks is unacceptable due to undue wear, mechanical failure, or loosening of parts, the test hooks shall be rejected.

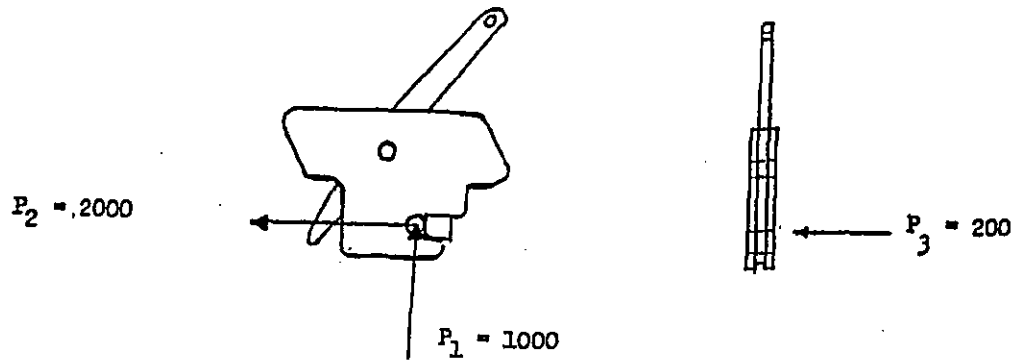
4.6.6 Static and repeat load tests.

4.6.6.1 Static tests. Static tests shall be conducted on the sample hooks with loads shown applied simultaneously as follows.

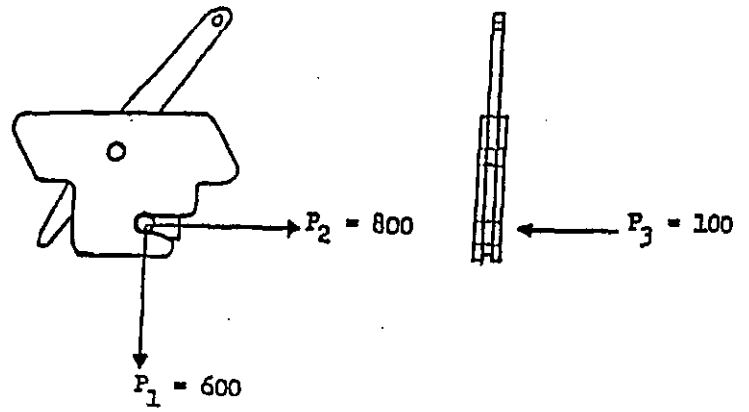
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4.6.6.1.1 Thrust hook (NAF Drawing 400095). Loads shall be applied as shown.

a. Test no. 1:



b. Test no. 2:



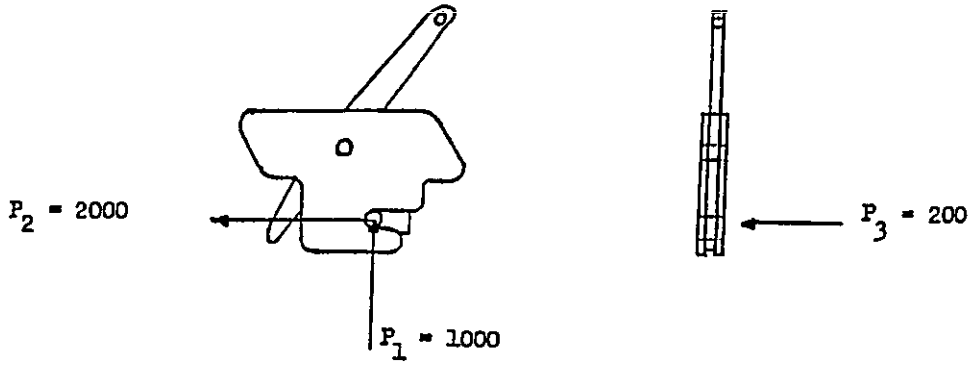
The above loads are limit loads. Tests shall be run to proof loads (1.10 x limit load) and there shall be no permanent deformation causing malfunctioning of any part of the hook below the proof load.

4.6.6.2 Repeat load tests. Repeat load tests shall be conducted on the sample hook as follows:

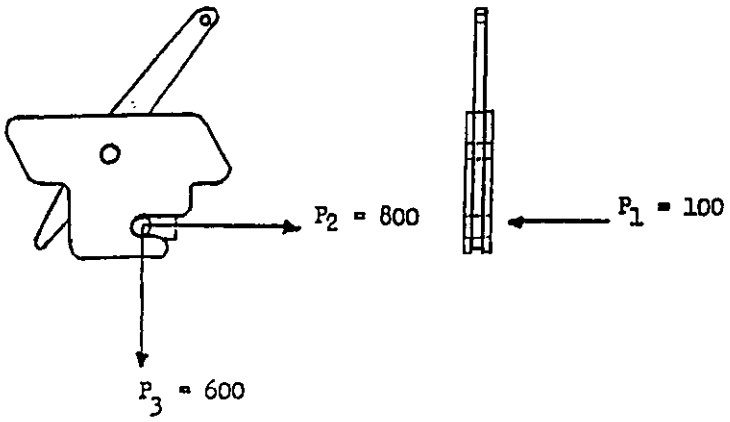
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4.6.6.2.1 Thrust hook (NAF Drawing 400095).

a. Test no. 3:



b. Test no. 4:



The repeat load tests are cycled from zero to the specified load and back to zero. There shall be no permanent deformation causing malfunctioning of parts below 2000 cycles.

4.6.7 Destruction tests.

4.6.7.1 Static destruction tests.

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4.6.7.1.1 Thrust hook (NAF Drawing 400095).

a. Loads shall be applied in the same manner as indicated in 4.6.6.1.1a and 4.6.6.1.1b. Failure shall not occur below values of:

P_2 = 3000 lbs P_1 = 1500 P_3 - 300

4.6.7.2 Repeat load destruction tests.

4.6.7.2.1 Thrust hook (NAF Drawing 400095).

a. Magnitude and direction of load to be as shown in 4.6.6.2.1a and 4.6.6.2.1b. Failure shall not occur below 25,000 cycles for first article testing, 15,000 cycles for production lot acceptance but to be continued to 25,000 cycles or failure whichever occurs first for quality control purposes.

4.7 Rejection. The failure of any sample hook to pass the quality conformance inspections and tests outlined in 4.4 shall be cause for rejection of all hooks represented by that sample, subject to retests.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or C as specified by the acquiring activity (see 6.2g).

* 5.1.1 Level A. Each thrust hook shall be individually preserved in accordance with MIL-P-116, Method III. The hook shall be placed within a transparent film bag of 0.0015 inch minimum gauge and heat sealed. The bag material shall conform to L-P-378 or the commercial equivalent. Each bagged hook shall be adequately cushioned, material conforming to A-A-1898, and enclosed within an individual close fitting fiberboard box conforming to PPP-B-636, Style OPF, Class Domestic, Grade 125; closure shall be accomplished in accordance with applicable procedures in the appendix.

* 5.1.2 Level C. Each hook shall be preserved in a manner that will prevent deterioration and damage during transit from the supply source to the first receiving activity. The contractor's commercial practice may be utilized when it conforms to the criteria stated herein.

* 5.2 Packing. Packing shall be level A, B, or C as specified by the acquiring activity (see 6.2g). Insofar as practicable, shipping containers shall effect a close fit, contain identical quantities, and be of uniform dimensional configuration.

* 5.2.1 Level A. Hooks, preserved as specified in 5.1.1, shall be packed for shipment in fiberboard containers conforming to PPP-B-636, Class Weather resistant, Grade V3s/V3c or PPP-B-640, Class 2, Style A. Each container shall be fitted with a waterproof liner conforming to MIL-L-10547. The gross weight of each packed container shall not exceed 70 pounds for the PPP-B-636 container nor 150 pounds for the PPP-B-640 container. The closure, sealing, and reinforcing (banding) of the containers shall be accomplished in accordance with the procedure specified in the appendix of the applicable container specification.

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* 5.2.2 Level B. Hooks, preserved as specified in 5.1.1, shall be packed for shipment in fiberboard containers conforming to PPP-B-636, Class Domestic, Grade 275 or PPP-B-640, Class 1, Style A or B. The gross weight of each packed container shall not exceed 90 pounds for the PPP-B-636 containers nor 200 pounds for PPP-B-640 containers. The closure and reinforcing (banding) of the containers shall be accomplished in accordance with the procedure specified in the appendix of the applicable container specification.

* 5.2.3 Level C. Hooks, preserved as specified in 5.1.2, shall be packed for shipment in a manner that will afford protection against damage during transit from the supply source to the first receiving activity. The pack shall, as a minimum, comply with the rules and regulations applicable to the mode of transportation utilized.

* 5.3 Marking. Each unit and shipping container shall be marked in accordance with MIL-STD-129 (see 6.2g). In addition, each unit pack shall be durably and legibly marked or labeled with the following information:

JATO hook - thrust
1000 pounds nominal thrust
Specification MIL-H-21660A(AS)
Preserved (date):
Domestic or overseas packed (as applicable)
Contract or order no.
Name of manufacturer
Name of contractor (if different from manufacturer)

* 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The thrust hooks covered by this specification are intended for use on certain aircraft to attach the JATO bottles.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article is required (see 3.1).

d. Whether first article inspection is waived (see 6.4).

e. Name and address of the inspection laboratory.

f. Hooks subjected to destructive tests should not be shipped as part of the acquisition quantity.

g. Applicable levels of preservation (see 5.1), packing (see 5.2), and marking requirements (see 5.3).

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h. Items of data required (see 6.3).

* 6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3.1	DI-NDTI-80809A	Test/inspection reports	----
4.3.1, 4.5.2, 4.5.3	DI-MISC-80678	Certification/data report	----

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

* 6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

* 6.5 Subject term (key word) listing.

Bottle
Limit load
Magnaflux inspection
Material failure
Vibration

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - AS
(Project 1340-N872)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-21660A(AS)	92/09/30

3. DOCUMENT TITLE: HOOK, THRUST, JATO, 1000 POUNDS NOMINAL THRUST

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488	(2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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